CPO experiment plan — mixed unary+pairwise regime

Implementation guide

CPO project

June 1, 2026

Abstract

This document specifies two main experiments and three follow-up experiments on the *mixed* regime of CPO, where unary single-label data and pairwise comparison data are combined within a single training objective. For each experiment: **Goal**, **Protocol**, **Figure specification**, **Expected outcome**.

1 Common setup

The world, policy, reference variant, and unary loss are unchanged from the base-rate heterogeneity experiment plan. Reuse your previous code. The additions here are: a pairwise data stream, a pairwise loss, and a mixing coefficient α.

World. Prompts $x \in \{1, \dots, P\}$ with $P = 8$; responses $y \in \{1, \dots, R\}$ with $R = 16$; fixed latent quality $q(x, y) \in [0, 1]$ drawn uniformly. Annotator clusters: Alice ($\tau_A = 0.25$, lenient) and Bob ($\tau_B = 0.75$, strict), with $\pi_A = 0.9$ unless otherwise stated. Label noise $\epsilon = 0.05$.

Unary stream. Each event:

1. Draw $x \sim \text{Uniform}(P)$, $y \sim \text{Uniform}(R)$, $c \sim \text{Cat}(\pi_A, 1 - \pi_A)$.
2. Label $s = \mathbb{1}\{q(x, y) > \tau_c\} \oplus \text{Bernoulli}(\epsilon)$.

Pairwise stream. Each event:

1. Draw $x \sim \text{Uniform}(P)$, two distinct $y_a, y_b \sim \text{Uniform}(R)$.
2. Set $y_w = \arg\max_{y \in \{y_a, y_b\}} q(x, y)$ (the true winner; all annotators agree on pairwise rankings in this world, since threshold heterogeneity does not affect ordering).
3. With probability $\epsilon_{\text{pair}} = 0.05$, swap y_w and y_ℓ (pairwise label noise).

Annotation budget. A pair label costs 2 unary labels of labeller effort (the annotator examines two responses). Hold *total effort* constant at $E = 256$ labels-equivalent per training step. If $f \in [0, 1]$ is the fraction of effort spent on pairs:

$$N_{\text{unary}} = \lfloor (1 - f)E \rfloor, \qquad N_{\text{pair}} = \lfloor fE/2 \rfloor.$$

Loss. For each minibatch:

$$\mathcal{L}^{\text{CPO}}_{\text{unary}} = \frac{1}{N_{\text{unary}}} \sum_{i=1}^{N_{\text{unary}}} \big[1 - \sigma(\beta\, m_i)\big], \quad m_i = \begin{cases} r_i - z_{c_i} & s_i = D \\ z_{c_i} - r_i & s_i = U \end{cases}$$

$$\mathcal{L}^{\text{DPO}}_{\text{pair}} = -\frac{1}{N_{\text{pair}}} \sum_{j=1}^{N_{\text{pair}}} \log \sigma\big(\beta\,(r_{w_j} - r_{\ell_j})\big)$$

$$\mathcal{L} = (1-\alpha)\,\mathcal{L}^{\text{CPO}}_{\text{unary}} + \alpha\,\mathcal{L}^{\text{DPO}}_{\text{pair}}.$$

The cluster references z_k are estimated by EMA over U-labelled unary samples only (the pairwise stream does not produce s labels):

$$z_k \leftarrow (1-\rho)\,z_k + \rho \cdot \frac{1}{|B_{k,U}|} \sum_{i \in B_{k,U}} r_i,$$

with $B_{k,U} = \{i : c_i = k, s_i = U\}$ over the unary minibatch. EMA rate $\rho = 0.1$, slope $\beta = 1$ unless otherwise stated.

Important edge cases. If $\alpha = 0$, ignore the pair stream entirely (do not sample pairs). If $\alpha = 1$, ignore the unary stream entirely. If a method's data stream is empty (e.g., KTO with $f > 0$, or DPO with $f < 1$), mark that configuration as N/A and do not train.

Policy and optimisation. Tabular cluster-blind policy $\theta \in \mathbb{R}^{P \times R}$, initialised to zero. Reference $\pi_{\text{ref}}(y \mid x) = 1/R$. Adam, learning rate 0.15, 500 training steps, evaluation every 25 steps.

Methods to compare.

- KTO: $\alpha = 0$, global z, all effort on unary.
- CPO ($\alpha = 0$): per-cluster z_k, all effort on unary.
- CPO ($\alpha = 0.5$): per-cluster z_k, mixed loss.
- DPO: $\alpha = 1$, all effort on pairs.

Evaluation. Closed-form expected true quality:

$$\mathbb{E}[q] = \frac{1}{P} \sum_x \sum_y \pi_\theta(y \mid x)\, q(x, y).$$

Seeds. Run each configuration with $n_{\text{seeds}} \in \{2, 3\}$ independent seeds (each regenerating the world, the data stream, and the policy initialisation).

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2 Experiment 1 — budget sweep

Goal

For each method, identify the allocation of labeller effort between unary and pairwise streams that maximises final $\mathbb{E}[q]$. The substantive question: *is there a regime where the mixed loss strictly dominates both pure unary and pure pairwise methods?*

Protocol

Sweep variable. The fraction of effort spent on pairs: $f \in \{0,\ 0.125,\ 0.25,\ 0.5,\ 0.75,\ 1.0\}$.

Run grid. 6 budgets $\times$ 4 methods $\times$ 2 seeds = 48 runs (minus degenerate cells; see edge cases).

What to log. Final $\mathbb{E}[q]$ per run.

Figure specification

A single 1×1 line plot, total size $\approx 8 \times 4.5$ inches.

- x-axis: f (fraction of effort spent on pairs), linear scale.
- y-axis: final $\mathbb{E}[q]$.
- Four curves, one per method, with ± 1 s.d. error bars over seeds.
- Title: "Budget sweep: when does mixing pay off?"
- Subtitle: total effort = 256 labels/step, $\pi_A = 0.9$.

Skip plotting points where a method's data stream is empty.

Expected outcome

Expected outcome

Mixed CPO ($\alpha = 0.5$) **dominates everywhere both streams are present**, by roughly 4–5 points over the better of DPO and pure CPO. DPO is approximately flat across f (a single ceiling determined by pair-noise). Pure CPO ($\alpha = 0$) degrades as f grows beyond $\sim$0.5 because it cannot use the pair stream and the unary stream becomes data-starved. KTO remains far below everyone else.

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3 Experiment 2 — α-sweep at fixed mixed budget

Goal

Isolate the mixing question from the budget question. With a fixed mixed budget ($N_{\text{unary}} = 128$, $N_{\text{pair}} = 64$), find the α that maximises $\mathbb{E}[q]$ and verify it is interior to $[0, 1]$.

Protocol

Sweep variable. $\alpha \in \{0.0, 0.1, 0.25, 0.5, 0.75, 0.9, 1.0\}$.

Fixed parameters. $N_{\text{unary}} = 128$, $N_{\text{pair}} = 64$ (so total effort $= 128 + 2 \cdot 64 = 256$). $\pi_A = 0.9$.

Edge cases. At $\alpha = 0$, only the unary stream is used; at $\alpha = 1$, only the pair stream is used. (The losses with zero weight have no effect; do not bother sampling that batch.)

Run grid. 7 values of $\alpha \times$ 2–3 seeds.

Figure specification

A 1×1 line plot, $\approx 7 \times 4.5$ inches.

- x-axis: α, linear scale on $[0, 1]$.
- y-axis: final $\mathbb{E}[q]$.
- One curve with ± 1 s.d. error bars; mark the argmax α^* with a vertical dashed line.
- Annotate endpoints: "CPO (unary only)" at $\alpha = 0$, "DPO (pairs only)" at $\alpha = 1$.

Expected outcome

Expected outcome

The curve has a clear **interior maximum** at $\alpha^* \in [0.25, 0.5]$, with the peak value 3–5 points above both endpoints. The endpoint values ($\alpha = 0$ and $\alpha = 1$) are statistically indistinguishable from each other. The curve is **broad and flat in the interior**: any $\alpha \in [0.1, 0.5]$ recovers nearly all the benefit, so exact tuning is not critical.

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4 Follow-up A — π_A sweep at fixed $\alpha = 0.5$

Goal

Combine the mixing axis with the heterogeneity axis. The previous π_A sweep (in the unary-only deliverable) showed an inverted-U in $\Delta_{\text{CPO}-\text{KTO}}$ peaking at moderate imbalance. The question here is whether mixed CPO ($\alpha = 0.5$) inherits this inverted-U, or whether the pairwise component (which is structurally robust to threshold heterogeneity) anchors the mixed objective and makes its π_A-trajectory flat.

Protocol

Sweep variable. $\pi_A \in \{0.5, 0.7, 0.85, 0.95, 0.99\}$.

Methods. KTO, CPO ($\alpha = 0$), mixed CPO ($\alpha = 0.5$) (budget split $N_{\text{unary}} = 128$, $N_{\text{pair}} = 64$), DPO. Use the same budgets across methods to keep effort comparable.

Run grid. 5 π_A values $\times$ 4 methods $\times$ 3 seeds.

Figure specification

A 1×2 panel figure, $\approx 11.5 \times 4$ inches.

- **Left:** x-axis π_A, y-axis final $\mathbb{E}[q]$, four curves, error bars.
- **Right:** same x-axis, y-axis difference vs. KTO ($\Delta = \mathbb{E}[q] - \mathbb{E}[q]_{\text{KTO}}$), three curves (CPO, mixed, DPO), horizontal line at $\Delta = 0$.

Expected outcome

Expected outcome

Mixed CPO ($\alpha = 0.5$) **is flat across π_A, above DPO by 4–5 points throughout.** Concretely:

- Mixed CPO: roughly 0.94 at every $\pi_A \in [0.5, 0.99]$.
- DPO: roughly 0.89 at every π_A (recovers the pi_A-flatness we already saw in the pure pairwise experiment).
- Pure CPO ($\alpha = 0$): inverted-U from $\sim$0.88 at $\pi_A = 0.5$, peaking around 0.86 at $\pi_A = 0.85$, then collapsing to $\sim$0.62 at $\pi_A = 0.99$.
- KTO: monotone degradation from $\sim$0.87 to $\sim$0.61.

Pedagogical takeaway: the pairwise stream in the mixed objective acts as an anchor against base-rate heterogeneity. Mixing *robustifies* CPO along the same dimension CPO was originally designed to fix.

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5 Follow-up B — optimal α^* as a function of pair budget

Goal

The optimal mixing coefficient should depend on how much pairwise data is available: more pairs $\Rightarrow$ higher α^* (rely more on the rich pairwise signal). With a small pair budget, the unary stream dominates and α^* should be small but nonzero. This produces a clean characterisation: α^* as a function of N_{pair}.

Protocol

Fixed parameters. $N_{\text{unary}} = 128$ per step (this is held *constant* so the unary stream's data volume is comparable across budgets). $\pi_A = 0.9$.

Pair budget sweep. $N_{\text{pair}} \in \{4, 16, 64, 256, 1024\}$ per step. Note this sweep *does not* hold total labeller effort constant; the question is about marginal benefit of additional pair data.

For each N_{pair}: sweep $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1.0\}$, run 2 seeds, record final $\mathbb{E}[q]$, and extract $\alpha^*(N_{\text{pair}}) = \arg\max_\alpha \mathbb{E}[q]$.

Run grid. 5 budgets $\times$ 7 α values $\times$ 2 seeds = 70 runs.

Figure specification

A 1×2 panel figure, $\approx 11.5 \times 4$ inches.

- **Left:** a family of curves — x-axis α, y-axis $\mathbb{E}[q]$, one curve per N_{pair} value (colour- gradient from light to dark by budget). Shows how the entire α-curve shifts with the pair budget.
- **Right:** x-axis N_{pair} (log scale), y-axis α^*, single curve with markers at the swept budget values.

Expected outcome

Expected outcome

$\alpha^*(N_{\mathbf{pair}})$ is monotone increasing, asymptoting to 0 from above as $N_{\mathbf{pair}} \to 0$ and to 1 from below as $N_{\mathbf{pair}} \to \infty$. Approximate values: $\alpha^*(4) \in [0, 0.1]$, $\alpha^*(16) \approx 0.25$, $\alpha^*(64) \approx 0.5$ (matches Experiment 2), $\alpha^*(256) \approx 0.75$, $\alpha^*(1024) \in [0.9, 1.0]$.
Conceptual handle: α^* is implicitly choosing a precision-weighted combination of two estimators. The unary loss has low per-sample information (binary thresholded) but high count; the pairwise loss has higher per-sample information (Bradley–Terry comparison) but lower count when pairs are scarce. As N_{pair} grows, the precision of the pairwise estimator grows roughly linearly, pulling α^* toward 1.
Side prediction: The peak $\mathbb{E}[q]$ at α^* is monotone nondecreasing in N_{pair}. More pair data is never worse.

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6 Follow-up C — ablation: per-cluster z_k vs. mixing

Goal

Our mixed-CPO baseline uses two ingredients: (i) per-cluster z_k references for the unary part, and (ii) mixing with pairs. Which ingredient is doing the work? A 2×2 design disentangles them. The most interesting hypothesis to test: *does the per-cluster z_k benefit shrink when the pairwise term is added?* If so, the pairwise term is partially absorbing the calibration role.

Protocol

Fixed parameters. $N_{\text{unary}} = 128$, $N_{\text{pair}} = 64$. $\pi_A = 0.9$.

2×2 design.

	Global z (KTO-style)	**Per-cluster** z_k (CPO-style)
$\alpha = 0$ (no pair term)	cell A (= KTO)	cell B (= pure CPO)
$\alpha = 0.5$ (mixed)	cell C ("mixed KTO")	cell D (= mixed CPO)

The crucial cell is C. "Mixed KTO" is not previously trained: it uses a single global z for the unary part but mixes with the DPO pairwise loss. This isolates whether the mixing benefit comes from the pair term itself (in which case C $\approx$ D) or from the combination of pair term and per-cluster calibration (in which case D $\gg$ C).

Run grid. 4 cells $\times$ 3 seeds.

Figure specification

A 1×1 grouped bar chart, $\approx 7 \times 4.5$ inches.

- Two groups along x: $\alpha = 0$ and $\alpha = 0.5$.
- Within each group: two bars (global z, per-cluster z_k).
- Error bars: ± 1 s.d. over seeds.
- Annotate each bar with its numerical value.

Expected outcome

Expected outcome

The z_k benefit **shrinks dramatically when mixing is added.** Approximate cell values:

	Global z	Per-cluster z_k	
$\alpha = 0$	A: $\sim$0.72	B: $\sim$0.87	($\Delta = +0.15$)
$\alpha = 0.5$	C: $\sim$0.92	D: $\sim$0.94	($\Delta = +0.02$)

Pedagogical takeaway: most of CPO's value in the *unary-only* regime comes from the per-cluster references (B $\gg$ A). Most of the value in the *mixed* regime comes from the mixing itself (C $\gg$ A, even without per-cluster z_k). The two mechanisms are partial substitutes: once the pairwise loss is doing the heavy lifting on ranking, the calibration mechanism matters less but still helps modestly.

Important alternative scenario the intern should consider. If the pair budget were smaller (e.g., $N_{\text{pair}} = 8$ instead of 64), the pairwise loss would not be strong enough to absorb the calibration role, and the z_k gap at $\alpha = 0.5$ would remain large (similar to the gap at $\alpha = 0$). Worth running this secondary configuration to confirm the dependence.
